

Full Title of Article

This Title Has A Line Break

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Abstract

This is the abstract for this article.

Keywords: List of keywords

or the L^AT_EX Community Forum (<http://www.latex-community.org/forum/>).

NOTE:

This is an numbered theorem-like environment that was defined in this document’s preamble.

1. Introduction

This is a sample article that uses the `jmlr` class with the `wcp` class option. Please follow the guidelines in this sample document as it can help to reduce complications when combining the articles into a book. Please avoid using obsolete commands, such as `\rm`, and obsolete packages, such as `epsfig`.¹ Some packages that are known to cause problems for the production editing process are checked for by the `jmlr` class and will generate an error. (If you want to know more about the production editing process, have a look at the video tutorials for the production editors at <http://www.dickimaw-books.com/software/makejmlrbookgui/videos/>.)

Please also ensure that your document will compile with PDF^LA^TE^X. If you have an error message that’s puzzling you, first check for it at the UK TUG FAQ <https://texfaq.org/FAQ-man-latex>.

If that doesn’t help, create a minimal working example (see <https://www.dickimaw-books.com/latex/minexample/>) and post to somewhere like T_EX on StackExchange (<http://tex.stackexchange.com/>)

1.1. Sub-sections

Sub-sections are produced using `\subsection`.

1.1.1. SUB-SUB-SECTIONS

Sub-sub-sections are produced using `\subsubsection`.

Sub-sub-sub-sections Sub-sub-sub-sections are produced using `\paragraph`. These are unnumbered with a running head.

Sub-sub-sub-sub-sections Sub-sub-sub-sub-sections are produced using `\subparagraph`. These are unnumbered with a running head.

2. Cross-Referencing

Always use `\label` and `\ref` (or one of the commands described below) when cross-referencing. For example, the next section is Section 3 but you can also refer to it using Section 3. The `jmlr` class provides some convenient cross-referencing

1. See <http://www.ctan.org/pkg/l2tabu>

commands: `\sectionref`, `\equationref`, `\tableref`, `\figureref`, `\algorithmref`, `\theoremref`, `\lemmaref`, `\remarkref`, `\corollaryref`, `\definitionref`, `\conjectureref`, `\axiomref`, `\exempleref` and `\appendixref`. The argument of these commands may either be a single label or a comma-separated list of labels. Examples:

Referencing sections: Section 3 or Sections 1 and 3 or Sections 1, 3, 5.1 and 5.2.

Referencing equations: Equation (1) or Equations (1) and (3) or Equations (1), (2), (3) and (4).

Referencing tables: Table 1 or Tables 1 and 2 or Tables 1, 2 and 3.

Referencing figures: Figure 1 or Figures 1 and 2 or Figures 1, 2 and 3 or Figures 3(a) and 3(b).

Referencing algorithms: Algorithm 1 or Algorithms 1 and 2 or Algorithms 1, 2 and 3.

Referencing theorem-like environments: Theorem 1, Lemma 2, Remark 3, Corollary 4, Definition 5, Conjecture 6, Axiom 7 and Example 1.

Referencing appendices: Appendix A or Appendices A and B.

3. Equations

The `jmlr` class loads the `amsmath` package, so you can use any of the commands and environments defined there. (See the `amsmath` documentation for further details.²)

Unnumbered single-lined equations should be displayed using `\[` and `\]`. For example:

$$E = mc^2$$

or you can use the `displaymath` environment:

$$E = mc^2$$

2. Either `texdoc amsmath` or <http://www.ctan.org/pkg/amsmath>

Numbered single-line equations should be displayed using the `equation` environment. For example:

$$\cos^2 \theta + \sin^2 \theta \equiv 1 \quad (1)$$

This can be referenced using `\label` and `\equationref`. For example, Equation (1).

Multi-lined numbered equations should be displayed using the `align` environment.³ For example:

$$f(x) = x^2 + x \quad (2)$$

$$f'(x) = 2x + 1 \quad (3)$$

Unnumbered multi-lined equations can be displayed using the `align*` environment. For example:

$$\begin{aligned} f(x) &= (x+1)(x-1) \\ &= x^2 - 1 \end{aligned}$$

If you want to mix numbered with unnumbered lines use the `align` environment and suppress unwanted line numbers with `\nonumber`. For example:

$$\begin{aligned} y &= x^2 + 3x - 2x + 1 \\ &= x^2 + x + 1 \end{aligned} \quad (4)$$

An equation that is too long to fit on a single line can be displayed using the `split` environment. Text can be embedded in an equation using `\text` or `\intertext` (as used in Theorem 1). See the `amsmath` documentation for further details.

3.1. Operator Names

Predefined operator names are listed in Table 1. For additional operators, either use `\operatorname`, for example `\operatorname{var}(X)` or declare it with `\DeclareMathOperator`, for example

3. For reasons why you shouldn't use the obsolete `eqnarray` environment, see Lars Madsen, *Avoid eqnarray!* TUGboat 33(1):21–25, 2012.

`\DeclareMathOperator{\var}{var}`

and then use this new command. If you want limits that go above and below the operator (like `\sum`) use the starred versions (`\operatorname*` or `\DeclareMathOperator*`).

4. Vectors and Sets

Vectors should be typeset using `\vec`. For example $\mathbf{x}$. (The original version of `\vec` can also be accessed using `\orgvec`, for example $\vec{x}$.) The `jmlr` class also provides `\set` to typeset a set. For example $\mathcal{S}$.

5. Floats

Floats, such as figures, tables and algorithms, are moving objects and are supposed to float to the nearest convenient location. Please don't force them to go in a particular place. In general it's best to use the `htbp` specifier and don't put the figure or table in the middle of a paragraph (that is make sure there's a paragraph break above and below the float). Floats are supposed to have a little extra space above and below them to make them stand out from the rest of the text. This extra spacing is put in automatically and shouldn't need modifying.

If your article will later be reprinted in the Challenges for Machine Learning, please be aware that the CiML books use a different paper size, so if you want to resize any images use a scale relative to the line width (`\linewidth`), text width (`\textwidth`) or text height (`\textheight`).

To ensure consistency, please *don't* try changing the format of the caption by doing something like:

`\caption{\textit{A Sample Caption.}}`

or

`\caption{\em A Sample Caption.}`

You can, of course, change the font for individual words or phrases, for example:

`\caption{A Sample Caption With Some \emph{Emphasized`

5.1. Tables

Tables should go in the `table` environment. Within this environment use `\floatconts` (defined by `jmlr`) to set the caption correctly and center the table contents. The location of the caption depends on the `tablecaption` setting in the document class options.

If you want horizontal rules you can use the `booktabs` package which provides the commands `\toprule`, `\midrule` and `\bottomrule`. For example, see Table 3.

If you really want vertical lines as well, you can't use the `booktabs` commands as there'll be some unwanted gaps. Instead you can use L^AT_EX's `\hline`, but the rows may appear a bit cramped. You can add extra space above or below a row using `\abovestrut` and `\belowstrut`. For example, see Table 4. However, you might want to read the `booktabs` documentation regarding the use of vertical lines.

If you want to align numbers on their decimal point, you can use the `siunitx` package. For example, see Table 5. For further details see the `siunitx` documentation⁴.

If the table is too wide, you can adjust the inter-column spacing by changing the value of `\tabcolsep`. For example:

`\setlength{\tabcolsep}{3pt}`

If the table is very wide but not very long, you can use the `sidewaystable` environment defined in the `rotating` package (so use `\usepackage{rotating}`). If the table is too long to fit on a page, you can use the `longtable` environment defined in the `longtable` package (so use `\usepackage{longtable}`).

4. Either `texdoc siunitx` or <http://www.ctan.org/pkg/siunitx>

Table 1: Predefined Operator Names (taken from `amsmath` documentation)

<code>\arccos</code>	<code>arccos</code>	<code>\deg</code>	<code>deg</code>	<code>\lg</code>	<code>lg</code>	<code>\projlim</code>	<code>projlim</code>
<code>\arcsin</code>	<code>arcsin</code>	<code>\det</code>	<code>det</code>	<code>\lim</code>	<code>lim</code>	<code>\sec</code>	<code>sec</code>
<code>\arctan</code>	<code>arctan</code>	<code>\dim</code>	<code>dim</code>	<code>\liminf</code>	<code>liminf</code>	<code>\sin</code>	<code>sin</code>
<code>\arg</code>	<code>arg</code>	<code>\exp</code>	<code>exp</code>	<code>\limsup</code>	<code>limsup</code>	<code>\sinh</code>	<code>sinh</code>
<code>\cos</code>	<code>cos</code>	<code>\gcd</code>	<code>gcd</code>	<code>\ln</code>	<code>ln</code>	<code>\sup</code>	<code>sup</code>
<code>\cosh</code>	<code>cosh</code>	<code>\hom</code>	<code>hom</code>	<code>\log</code>	<code>log</code>	<code>\tan</code>	<code>tan</code>
<code>\cot</code>	<code>cot</code>	<code>\inf</code>	<code>inf</code>	<code>\max</code>	<code>max</code>	<code>\tanh</code>	<code>tanh</code>
<code>\coth</code>	<code>coth</code>	<code>\injlim</code>	<code>injlim</code>	<code>\min</code>	<code>min</code>		
<code>\csc</code>	<code>csc</code>	<code>\ker</code>	<code>ker</code>	<code>\Pr</code>	<code>Pr</code>		
		<code>\varlimsup</code>	$\overline{\lim}$	<code>\varinjlim</code>	$\varinjlim$		
		<code>\varliminf</code>	$\underline{\lim}$	<code>\varprojlim</code>	$\varprojlim$		

Table 2: An Example Table

Dataset	Result
Data1	0.12345
Data2	0.67890
Data3	0.54321
Data4	0.09876

Table 5: A Table With Numbers Aligned on the Decimal Point

Dataset	Result
Data1	0.123 45
Data2	10.6789
Data3	50.543
Data4	200.098 76

Table 3: A Table With Horizontal Lines

Dataset	Result
Data1	0.12345
Data2	0.67890
Data3	0.54321
Data4	0.09876

Table 4: A Table With Horizontal and Vertical Lines

Dataset	Result
Data1	0.12345
Data2	0.67890
Data3	0.54321
Data4	0.09876

5.2. Figures

Figures should go in the `figure` environment. Within this environment, use `\floatconts` to correctly position the caption and center the image. Use `\includegraphics` for external graphics files but omit the file extension. Do not use `\epsfig` or `\psfig`. If you want to scale the image, it's better to use a fraction of the line width rather than an explicit length. For example, see Figure 1.

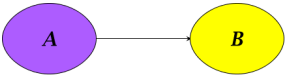


Figure 1: Example Image

If your image is made up of L^AT_EX code (for example, commands provided by the `pgf` package) you can include it using `\includeteximage` (defined by the `jmlr` class). This can be scaled and rotated in the same way as `\includegraphics`. For example, see Figure 2.

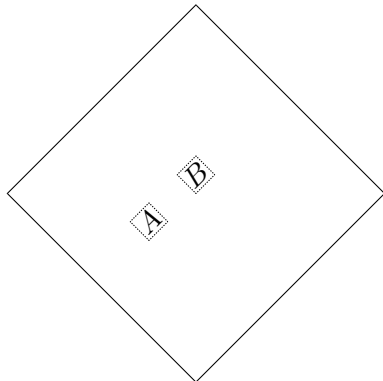


Figure 2: Image Created Using L^AT_EX Code

If the figure is too wide to fit on the page, you can use the `sidewaysfigure` environment defined in the `rotating` package.

Don't use `\graphicspath`.⁵ If the images are contained in a subdirectory, specify this when you include the image, for example `\includegraphics{figures/mypic}`.

5.2.1. SUB-FIGURES

Sub-figures can be created using `\subfigure`, which is defined by the `jmlr` class. The optional argument allows you to provide a subcaption. The label should be placed in the mandatory argument of `\subfigure`. You can reference the entire figure, for example Figure 3, or you can reference part of the figure using

5. This is specific to the `jmlr` class, not a general recommendation. The main file that generates the proceedings or the CiML book is typically in a different directory to the imported articles, so it modifies the graphics path when it imports an article.

`\figureref`, for example Figure 3(a). Alternatively you can reference the subfigure using `\subfigref`, for example (a) and (b) in Figure 3.

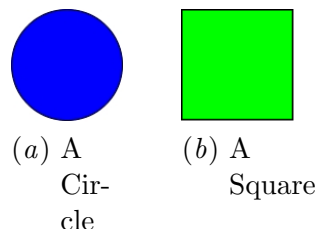


Figure 3: An Example With Sub-Figures.

By default, the sub-figures are aligned on the baseline. This can be changed using the second optional argument of `\subfigure`. This may be `t` (top), `c` (centered) or `b` (bottom). For example, the subfigures (a) and (b) in Figure 4 both have `[c]` as the second optional argument.

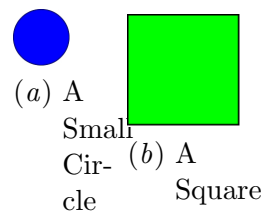


Figure 4: Another Example With Sub-Figures.

5.3. Sub-Tables

There is an analogous command `\subtable` for sub-tables. It has the same syntax as `\subfigure` described above. You can reference the table using `\tableref`, for example Table 6 or you can reference part of the table, for example Table 6(a). Alternatively you can reference the subtable using `\subtabref`, for example (a) and (b) in Table 6.

Table 6: An Example With Sub-Tables

(a)		(b)	
A	B	C	D
1	2	3	4
		5	6

By default, the sub-tables are aligned on the top. This can be changed using the second optional argument of `\subtable`. This may be `t` (top), `c` (centered) or `b` (bottom). For example, the sub-tables (a) and (b) in Table 7 both have `[c]` as the second optional argument.

Table 7: Another Example With Sub-Tables

(a)		(b)	
A	B	C	D
1	2	3	4
		5	6

5.4. Algorithms

Enumerated textual algorithms can be displayed using the `algorithm` environment. Within this environment, use `\caption` to set the caption and you can use an `enumerate` or nested `enumerate` environments. For example, see Algorithm 1. Note that algorithms float like figures and tables.

If you'd rather have the same numbering throughout the algorithm but still want the convenient indentation of nested `enumerate` environments, you can use the `enumerate*` environment provided by the `jmlr` class. For example, see Algorithm 2.

Pseudo code can be displayed using the `algorithm2e` environment. This is defined by the `algorithm2e` package (which is automatically loaded) so check the `algorithm2e`

Algorithm 1: The Gauss-Seidel Algorithm

1. For $k = 1$ to maximum number of iterations
 - (a) For $i = 1$ to n
 - i. $x_i^{(k)} = \frac{b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k)} - \sum_{j=i+1}^n a_{ij}x_j^{(k-1)}}{a_{ii}}$
 - ii. If $\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\| < \epsilon$, where ϵ is a specified stopping criteria, stop.
-

Algorithm 2: Moore's Shortest Path

Given a connected graph G , where the length of each edge is 1:

1. Set the label of vertex s to 0
 2. Set $i = 0$
 3. Locate all unlabelled vertices adjacent to a vertex labelled i and label them $i + 1$
 4. If vertex t has been labelled,

the shortest path can be found by backtracking, and the length is given by the label of t .

otherwise

increment i and return to step 3
-

documentation for further details.⁶ For an example, see Algorithm 3.

Algorithm 3: Computing Net Activation

Input: $x_1, \dots, x_n, w_1, \dots, w_n$

Output: y , the net activation

```

 $y \leftarrow 0;$ 
for  $i \leftarrow 1$  to  $n$  do
  |  $y \leftarrow y + w_i * x_i;$ 
end

```

6. Description Lists

The `jmlr` class also provides a description-like environment called `altdescription`. This has an argument that should be the widest label in the list. Compare:

add A method that adds two variables.

differentiate A method that differentiates a function.

with

add A method that adds two variables.

differentiate A method that differentiates a function.

7. Theorems, Lemmas etc

The following theorem-like environments are predefined by the `jmlr` class: `theorem`, `example`, `lemma`, `proposition`, `remark`, `corollary`, `definition`, `conjecture` and `axiom`. You can use the `proof` environment to display the proof if need be, as in Theorem 1.

Theorem 1 (Eigenvalue Powers) *If λ is an eigenvalue of B with eigenvector ξ , then*

6. Either `texdoc algorithm2e` or <http://www.ctan.org/pkg/algorithm2e>

λ^n is an eigenvalue of B^n with eigenvector ξ .

Proof *Let λ be an eigenvalue of B with eigenvector ξ , then*

$$B\xi = \lambda\xi$$

premultiply by B :

$$\begin{aligned}
 BB\xi &= B\lambda\xi \\
 \Rightarrow B^2\xi &= \lambda B\xi \\
 &= \lambda\lambda\xi && \text{since } B\xi = \lambda\xi \\
 &= \lambda^2\xi
 \end{aligned}$$

Therefore true for $n = 2$. Now assume true for $n = k$:

$$B^k\xi = \lambda^k\xi$$

premultiply by B :

$$\begin{aligned}
 BB^k\xi &= B\lambda^k\xi \\
 \Rightarrow B^{k+1}\xi &= \lambda^k B\xi \\
 &= \lambda^k\lambda\xi && \text{since } B\xi = \lambda\xi \\
 &= \lambda^{k+1}\xi
 \end{aligned}$$

Therefore true for $n = k + 1$. Therefore, by induction, true for all n . ■

Lemma 2 (A Sample Lemma) *This is a lemma.*

Remark 3 (A Sample Remark) *This is a remark.*

Corollary 4 (A Sample Corollary) *This is a corollary.*

Definition 5 (A Sample Definition) *This is a definition.*

Conjecture 6 (A Sample Conjecture) *This is a conjecture.*

Axiom 7 (A Sample Axiom) *This is an axiom.*

Example 1 (An Example) *This is an example.*

8. Color vs Grayscale

It's helpful if authors supply grayscale versions of their images in the event that the article is to be incorporated into a black and white printed book. With external PDF, PNG or JPG graphic files, you just need to supply a grayscale version of the file. For example, if the file is called `myimage.png`, then the gray version should be `myimage-gray.png` or `myimage-gray.pdf` or `myimage-gray.jpg`. You don't need to modify your code. The `jmlr` class checks for the existence of the grayscale version if it is print mode (provided you have used `\includegraphics` and haven't specified the file extension).

You can use `\ifprint` to determine which mode you are in. For example, in Figure 1, the purple ellipse represents an input and the yellow ellipse represents an output. Another example: **important text!**

You can use the class option `gray` to see how the document will appear in gray scale mode. **Colored text** will automatically be converted to gray scale in print mode.

The `jmlr` class loads the `xcolor` package, so you can also define your own colors. For example: **XYZ**.

The `xcolor` class is loaded with the `x11names` option, so you can use any of the x11 predefined colors (listed in the `xcolor` documentation⁷).

9. Citations and Bibliography

The `jmlr` class automatically loads `natbib` and automatically sets the bibliography style, so you don't need to use `\bibliographystyle`. This sample file has the citations defined in the accompanying BibTeX file `jmlr-sample.bib`. For a parenthetical citation use `\citep`. For exam-

ple (?). For a textual citation use `\citet`. For example ?. Both commands may take a comma-separated list, for example ??.

These commands have optional arguments and have a starred version. See the `natbib` documentation for further details.⁸

The bibliography is displayed using `\bibliography`.

Acknowledgments

Acknowledgements go here.

References

- I. Guyon and A. Elisseeff. An introduction to variable and feature selection. *JMLR*, 3:1157–1182, March 2003.
- I. Guyon, C. Aliferis, and A. Elisseeff. Causal feature selection. Technical report, Clopinet, 2007.

Appendix A. First Appendix

This is the first appendix.

Appendix B. Second Appendix

This is the second appendix.

7. either `texdoc xcolor` or <http://www.ctan.org/pkg/xcolor>

8. Either `texdoc natbib` or <http://www.ctan.org/pkg/natbib>